

Government College of Engineering, Karad

OE - AISDML

Project Presentation

Title : Surface Anomaly Detection

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Manufacturing Surface Anomaly Inspector

Deep Learning Based Multi-label Steel Surface Defect Detection

- **Dataset:** Steel Surface Defects — 2,249+ images, 10 defect classes
- **Models:** YOLOv8s
- RT-DETR (Based on the ResNet)
- **Goal:** Real-time automated inspection replacing manual quality control

PROBLEM STATEMENT

What Problem Are We Solving?

The Challenge

- Steel lines generate thousands of images per hour — manual inspection cannot keep pace
- Human inspection is slow, inconsistent, and costly
- Traditional image processing fails on subtle defects (inclusion, crease)
- A single missed defect causes product failure and financial loss

Our Goal

- Detect 10 steel surface defect types automatically
- Trained YOLOv8s and RT-DETR to identify the best performer
- Deploy as a real-time inspection dashboard on the production line

Steel Surface Defect Dataset

10 Defect Classes

Dataset Facts

- 2,249 grayscale images
- 3,550 bounding box annotations
- Resolution: $\sim 2048 \times 1000$ px.
- Images with their labels
- Format: Pascal VOC XML
- Grayscale images

ID	Class	Description
0	Crease	Thin curved lines
1	Crescent Gap	Curved edge cutout
2	Inclusion	Embedded dark spots
3	Oil Spot	Blob contamination
4	Punching Hole	Circular edge hole
5	Rolled Pit	Pressed depressions
6	Silk Spot	Hair-like scratches
7	Waist Folding	Horizontal folds
8	Water Spot	Diagonal streak
9	Welding Line	Edge weld shape change

Understanding the Raw Dataset

Image Characteristics

- ✓ All grayscale — hue/saturation augmentation is ineffective
- ✓ Large black region (20–70% of image) on left — camera artifact
- ✓ Brightness varies significantly across images and classes
- ✓ 3 camera zones produce distinct visual contexts

Bounding Box Statistics

- ✓ Width: 14–2047 px.
- ✓ Height: 16–999 px.
- ✓ Aspect ratio: 0.06–67.53 (dots to wide bands)
- ✓ Average defects per image: 1.56

Critical Finding: Severe Class Imbalance

Class	Annotations	Status
Silk Spot	892	Overrepresented
Oil Spot	569	High
Welding Line	506	Medium
Inclusion	344	Medium
Crease	74	Too Low
Rolled Pit	11	Critical

Fixing Class Imbalance via Augmentation

Strategy: Offline Augmentation

- Generated synthetic images for underrepresented classes
- Target: 500–700 annotations per class (from minimum 11)
- Only augmented images containing the target class
- Bounding boxes auto-adjusted per transformation

Augmentation Techniques

Technique	Purpose
Horizontal / Vertical Flip	Position generalization
Brightness / Contrast	Camera exposure variation
Random Rotation ($\pm 10^\circ$)	Orientation variation
Elastic Transform	Surface texture variation
Gaussian Noise	Sensor noise simulation
CLAHE	Local contrast boost
Grid Distortion	Surface deformation
Mosaic Augmentation	Tiny defect detection

Results After Balancing

Class	Before	After
Rolled Pit	11	543
Crease	47	<700
Silk Spot	892	<700
All Others	74–569	500–640
Total	2,449	6,476

Preparing Images for Training

Dataset Split



Step 1 — Format Conversion

- Pascal VOC XML → YOLO TXT (normalized 0–1)
- Filtered 2 extra classes (scratch, unknown)
- Fixed class ID ordering for deterministic sessions

Step 2 — CLAHE Enhancement

- Resizing Image
- Tile grid: 8×8
- Enhances local contrast on dark images
- Critical for Inclusion class (near-invisible spots)

Step 3 — Dataset Split

- Train 70%
- Val 20%
- Test 10%
- Split at image level — prevents data leakage

YOLOv8s: The Fast & Efficient Scanner

High-speed detection using a "Single-Shot" approach.



The Backbone (The Eyes)

Uses a C2f module to break down the image and find shapes like "dents" or "cracks" very quickly.



The Head (The Decision Maker)

Utilizes a Decoupled Head, meaning it identifies what the object is and where it is in two separate steps for better accuracy.



The Neck (The Context)

A PANet structure that mixes small details with large shapes so the model doesn't miss tiny defects.



Key Advantage: Anchor-Free

Predicts the center of a defect directly rather than guessing from pre-set anchor boxes, simplifying training.

RT-DETR: The Smart Transformer

Real-time accuracy using "Global Attention."



Hybrid Encoder

Combines CNN speed with Transformer's global attention to focus on important image regions efficiently.



Query Selection

Instead of exhaustive search, it intelligently selects 300 "queries" to pinpoint defect locations with precision.



NMS-Free

An "End-to-End" model that directly predicts correct bounding boxes, eliminating the need for non-maximum suppression (NMS).



Key Advantage

Excels at detecting tiny defects against complex backgrounds (e.g., Inclusion class) by analyzing the entire image context.

YOLOv8s vs. RT-DETR: Performance Showdown

In a head-to-head evaluation, YOLOv8s significantly outperformed RT-DETR across all primary detection metrics on our steel surface defect dataset.

Metric	RT-DETR	YOLOv8s	Difference
mAP50 (Mean Accuracy)	0.732	0.914	+18.2%
Precision (P)	0.742	0.903	+16.1%
Recall (R)	0.721	0.861	+14.0%
mAP50-95 (Localization)	0.371	0.586	+21.5%

While RT-DETR provides a modern transformer-based approach, our YOLOv8s model proved much more effective for this specific dataset of surface defects. It offers a more balanced performance, higher precision for critical safety defects like 'welding lines,' and superior handling of small, low-contrast defects like 'inclusions.'"

Model Improvements

Enhancement	Current	Target	Expected Benefit
500+ real Inclusion images	65% mAP	85%+ mAP	Fix the only weak class
1280px resolution training	640px input	1280px input	Better tiny defect detection
YOLOv8l or YOLOv8x	YOLOv8s (11M)	43M–68M params	Higher accuracy, slower inference
Ensemble YOLOv8s + RT-DETR	Single model	Combined predictions	CNN speed + Transformer context
Instance Segmentation (YOLOv8-seg)	BBoxes only	Pixel masks	Exact defect boundary & area
Defect severity scoring (1–10)	Binary detect/not detect	Severity rating	Prioritize repairs by severity

Additional Features

- **Real-Time Video Stream Processing**
Video pipeline for production lines
- **Automatic Defect Report Generation**
PDF per coil: count, type, location
- **Alert System Integration**
Alarm on critical defect above threshold

THANK YOU